

Regression in Practice

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```
library(ggplot2) # install.packages("ggplot2")
library(fixest)  # install.packages("fixest")
library(arrow)   # install.packages("arrow")
```

Attaching package: 'arrow'

The following object is masked from 'package:utils':

timestamp

Intro

This R day is going to have you apply different regression strategies presented in topic 04. An emphasis will be placed on interpreting and presenting regression results. For this assignment, I have assembled a dataset consisting of homes in Massachusetts.

There are a lot of variables in this dataset, so please see the separate codebook for details.

Because this data is large (almost 2 million rows), I am using the **parquet** file format. You can load this with the **read_parquet** function from the **arrow** package. You might have to change your working directory in RStudio to the folder where this **.qmd** file is located.

```
# setwd("04-Regression_in_Practice/R-day/")
df <- read_parquet("data/MA_parcels_sample.parquet")
head(df)
```

	fy	town	census_tract	use_code	lot_size_acres	building_value
1	2019	Southborough	25027741101	101	0.31	165500
2	2020	Southborough	25027741101	101	0.31	171500
3	2022	Southborough	25027741101	101	0.31	182900

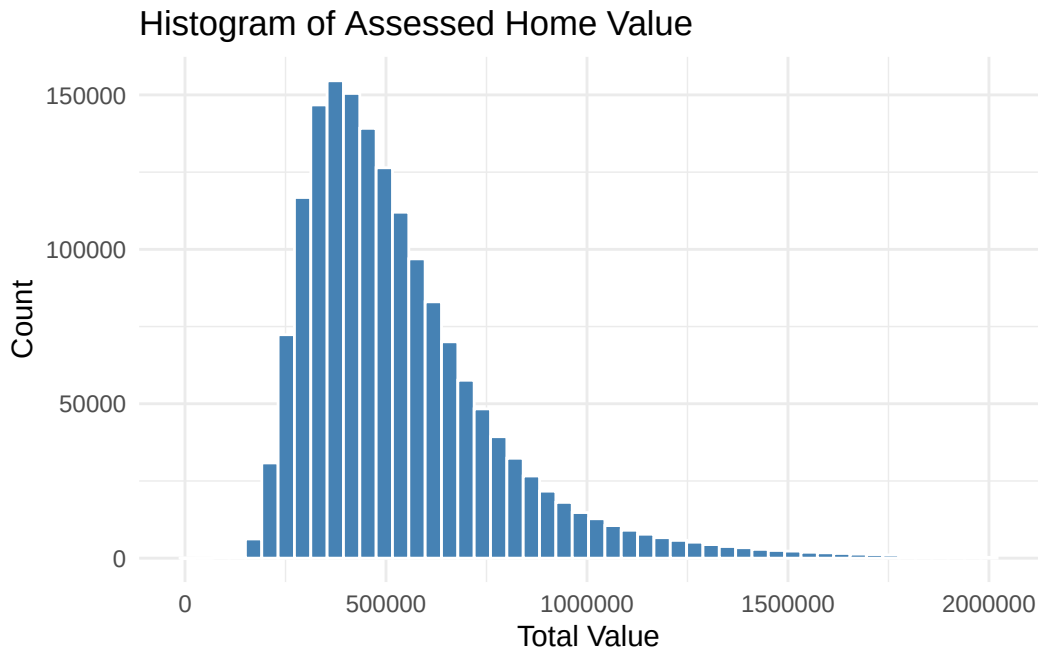
4	2023	Southborough	25027741101	101	0.31	169400
5	2013	Southborough	25027741101	101	1.34	344700
6	2019	Southborough	25027741101	101	1.34	414500
	land_value	other_value	total_value	last_sale_date	last_sale_year	
1	210400	7900	383800	1987-12-04	1987	
2	219400	8700	399600	1987-12-04	1987	
3	302700	8700	494300	1987-12-04	1987	
4	281600	8700	459700	2022-09-26	2022	
5	169200	0	513900	2008-03-31	2008	
6	206500	0	621000	2008-03-31	2008	
	last_sale_price	n_units	n_rooms	n_stories	style	year_built building_area
1	196500	1	6	1	OLD STYLE	1928 1837
2	196500	1	6	1	OLD STYLE	1928 1837
3	196500	1	6	1	OLD STYLE	1928 1837
4	435500	1	6	1	OLD STYLE	1928 1837
5	101	1	9	2	CL	2007 2817
6	101	1	9	2	COLONIAL	2007 2817
	residential_area	longitude	latitude	z_zoning_id	z_zoning_code	
1	1837	-71.52156	42.30717	Southborough --- 3	277RB	
2	1837	-71.52156	42.30717	Southborough --- 3	277RB	
3	1837	-71.52156	42.30717	Southborough --- 3	277RB	
4	1837	-71.52156	42.30717	Southborough --- 3	277RB	
5	2817	-71.52145	42.28848	Southborough --- 7	277RA	
6	2817	-71.52145	42.28848	Southborough --- 7	277RA	
	z_permit_2fam	z_permit_3_4fam	z_permit_5_19fam	z_permit_20_fam	z_dupac	
1	By Permit	By Permit	By Permit	By Permit	6	
2	By Permit	By Permit	By Permit	By Permit	6	
3	By Permit	By Permit	By Permit	By Permit	6	
4	By Permit	By Permit	By Permit	By Permit	6	
5	By Permit	By Permit	By Permit	By Permit	2	
6	By Permit	By Permit	By Permit	By Permit	2	
	z_max_height	z_mls_acres	z_residential	z_mixed	z_nonresidential	
1	35	0.3443526	TRUE	TRUE	FALSE	
2	35	0.3443526	TRUE	TRUE	FALSE	
3	35	0.3443526	TRUE	TRUE	FALSE	
4	35	0.3443526	TRUE	TRUE	FALSE	
5	35	1.0000000	TRUE	TRUE	FALSE	
6	35	1.0000000	TRUE	TRUE	FALSE	

Analysis

Basic exploration of variables

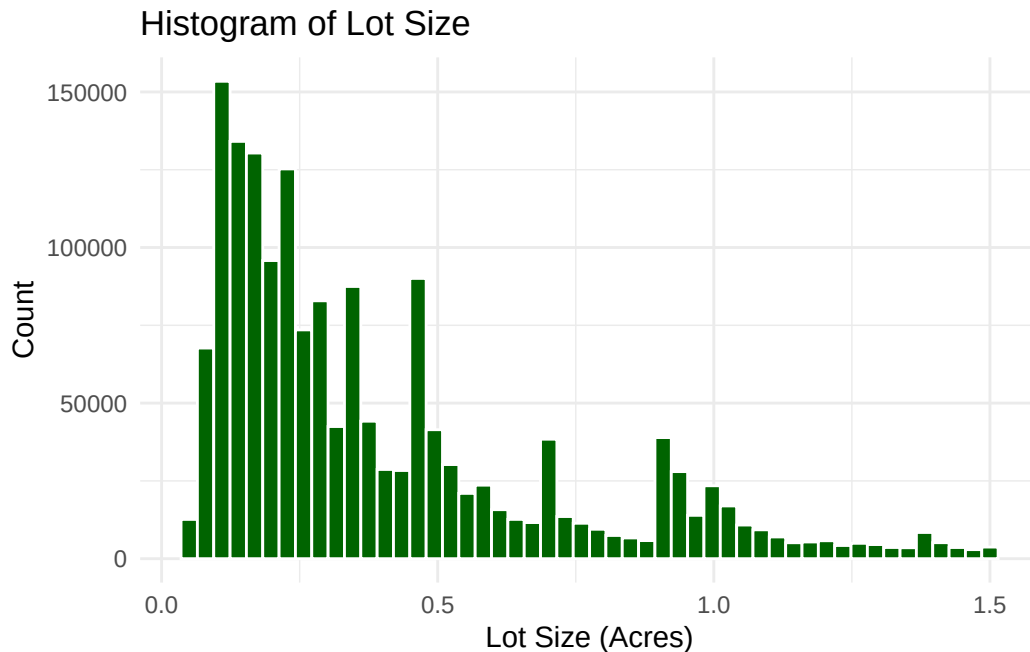
One of the main variables we will use in this analysis is the assessed value of the home (`total_value`) and the size of the land (`lot_size_acres`). Like good data analysts, let's create histograms of both variables. Comment on the distribution.

```
ggplot(df, aes(x = total_value)) +  
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +  
  labs(  
    title = "Histogram of Assessed Home Value",  
    x = "Total Value",  
    y = "Count"  
  ) +  
  theme_minimal()
```



```
ggplot(df, aes(x = lot_size_acres)) +  
  geom_histogram(bins = 50, fill = "darkgreen", color = "white") +  
  labs(  
    title = "Histogram of Lot Size",  
    x = "Lot Size (Acres)",
```

```
y = "Count"
) +
theme_minimal()
```



Answer: Both distributions are right-skewed, with a long tail of high values.

Now, let's look at a few explanatory variables. Use the `table` function to explore the distribution of `n_rooms` and make a histogram of `z_mls_acres`.

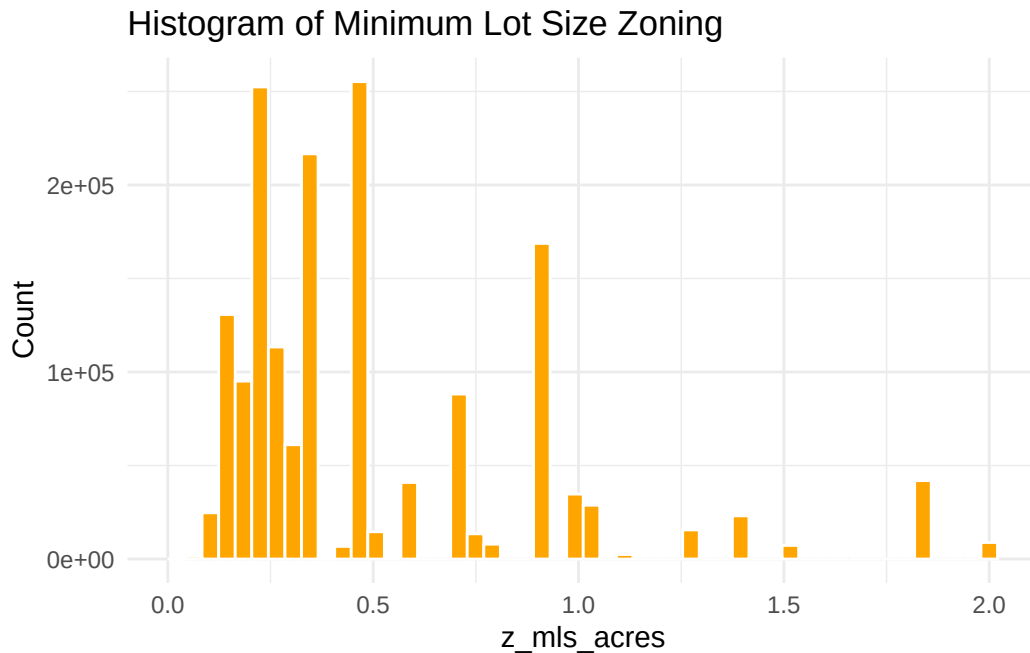
```
table(df$n_rooms)
```

```

  3    4    5    6    7    8    9   10
3980 41091 196951 441421 366967 337610 161447 105787

```

```
ggplot(df, aes(x = z_mls_acres)) +
  geom_histogram(bins = 50, fill = "orange", color = "white") +
  labs(
    title = "Histogram of Minimum Lot Size Zoning",
    x = "z_mls_acres",
    y = "Count"
  ) +
  theme_minimal()
```

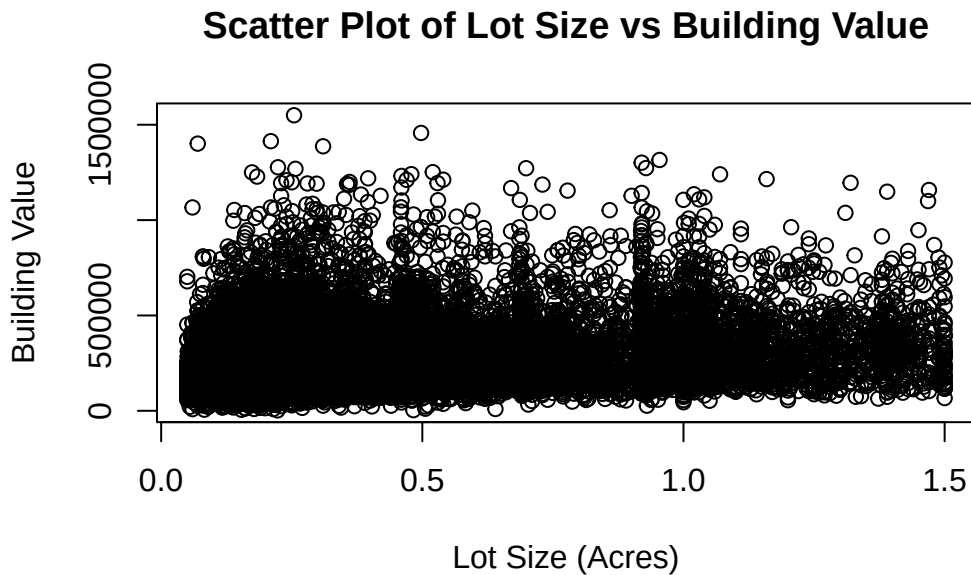


Answer: Also right-skewed, with a long tail of high values, with a large spike at 0.8 acres.

Relationship between lot size and building value

We are going to explore the relationship between the lot size and the value of the building. We will make a scatter plot to start our analysis. However, there is way too many points to plot in the dataset, so we will sample a random 10000 rows.

```
sample_df <- df[sample(1:nrow(df), 25000), ]
plot(sample_df$lot_size_acres, sample_df$building_value,
      xlab = "Lot Size (Acres)", ylab = "Building Value",
      main = "Scatter Plot of Lot Size vs Building Value")
```



Bivariate regression

First, regress building value on the lot size linearly using `feols` from the `fixest` package. Interpret the regression coefficient and comment on its statistical significance. Make sure to use `vcov = "HC1"` to get robust standard errors.

```
biv_mod <- feols(building_value ~ lot_size_acres, data = sample_df, vcov = "HC1")
summary(biv_mod)
```

```
OLS estimation, Dep. Var.: building_value
Observations: 25,000
Standard-errors: Heteroskedasticity-robust
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	216924.3	1419.63	152.8031	< 2.2e-16 ***
lot_size_acres	113666.0	3293.82	34.5089	< 2.2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 147,229.7   Adj. R2: 0.056182
```

Answer: A 1 acre increase in lot size is associated with a \$117,408.3 +/- \$3260.29 increase in building value. The result is statistically significant at the 1% level.

Forecasting

Now, let's try and show what our forecasted values will be. To do so, let's create a `data.frame` containing a grid of values of `lot_size_acres`. We can do that with `seq(0, 4, 0.01)`. Call this `pred_grid`. Create forecasts from your simple linear model using `predict(..., newdata = pred_grid)`. Store the forecasts back into the `pred_grid` data frame.

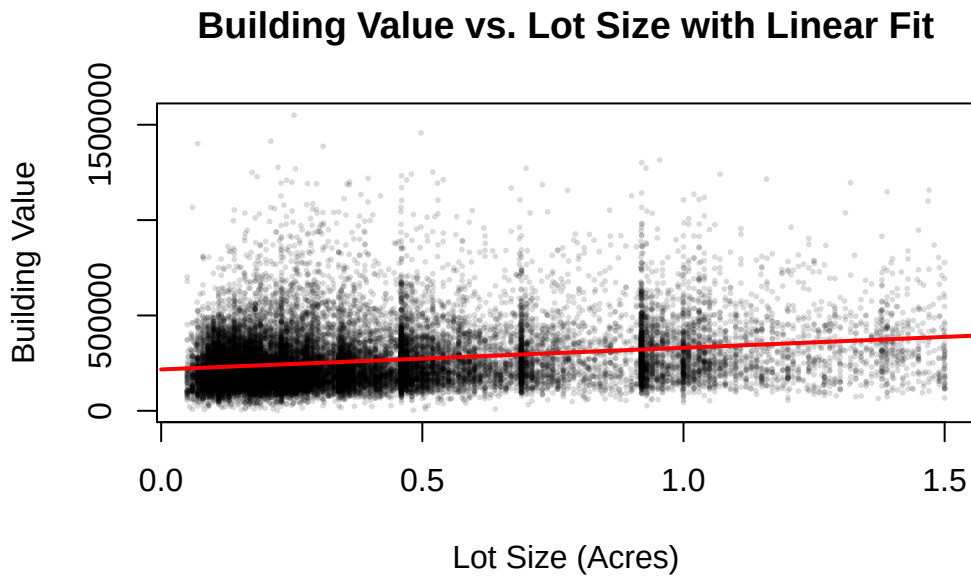
To create a plot, first copy your `plot` call above to create the sample scatterplot. Then, use the `lines` function to overlay your forecasted line of best fit. Give this line a distinct color and make the line thicker by adding the arguments `col = "red", lwd = 2`.

```
pred_grid <- data.frame(
  lot_size_acres = seq(0, 4, 0.01)
)

pred_grid$pred_building_value <- predict(biv_mod, newdata = pred_grid)

plot(
  sample_df$lot_size_acres,
  sample_df$building_value,
  pch = 16,
  cex = 0.4,
  col = rgb(0, 0, 0, 0.15),
  xlab = "Lot Size (Acres)",
  ylab = "Building Value",
  main = "Building Value vs. Lot Size with Linear Fit"
)

lines(
  pred_grid$lot_size_acres,
  pred_grid$pred_building_value,
  col = "red",
  lwd = 2
)
```



Forecasting with confidence intervals

Now, use your regression to predict the assessed value for a home on a 1.5 acre lot. Add the option `se.fit = TRUE` to your `predict` call to get a standard error for the predicted value. Present your result with a 95% confidence interval.

```
pred_15 <- predict(  
  biv_mod,  
  newdata = data.frame(lot_size_acres = 1.5),  
  se.fit = TRUE  
)  
  
predicted_value <- pred_15$fit  
se_pred <- pred_15$se.fit  
  
ci_lower <- predicted_value - 1.96 * se_pred  
ci_upper <- predicted_value + 1.96 * se_pred  
  
predicted_value
```

```
[1] 387423.3
```



```
ci_lower
```

```
[1] 379678.5
```

```
ci_upper
```

```
[1] 395168.1
```

Answer: Our predicted value for a home on a 1.5 acre lot is \$391,049.10, with a 95% confidence interval of [\$383,392.30, \$398,706.00].

Polynomial

Estimate a new model with a 4th-order polynomial of `lot_size_acres`. You can do this simply with the `poly` function. Forecast this model using `pred_grid` and create a new forecast plot similar to above. Add lines for both the linear fit (in red) and the polynomial fit (in blue)

```
poly_mod <- feols(building_value ~ poly(lot_size_acres, 4), data = df, vcov = "HC1")
summary(poly_mod)
```

OLS estimation, Dep. Var.: building_value

Observations: 1,655,254

Standard-errors: Heteroskedasticity-robust

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	262326.7	113.205	2317.27826	< 2.2e-16 ***
poly(lot_size_acres, 4)1	47546530.4	164486.277	289.06077	< 2.2e-16 ***
poly(lot_size_acres, 4)2	-5470262.5	160500.061	-34.08262	< 2.2e-16 ***
poly(lot_size_acres, 4)3	-1585829.9	162840.648	-9.73854	< 2.2e-16 ***
poly(lot_size_acres, 4)4	-2092704.8	160795.725	-13.01468	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 145,645.2 Adj. R2: 0.061412

```
pred_grid$pred_poly <- predict(poly_mod, newdata = pred_grid)
```

```
plot(
  sample_df$lot_size_acres,
  sample_df$building_value,
  pch = 16,
```

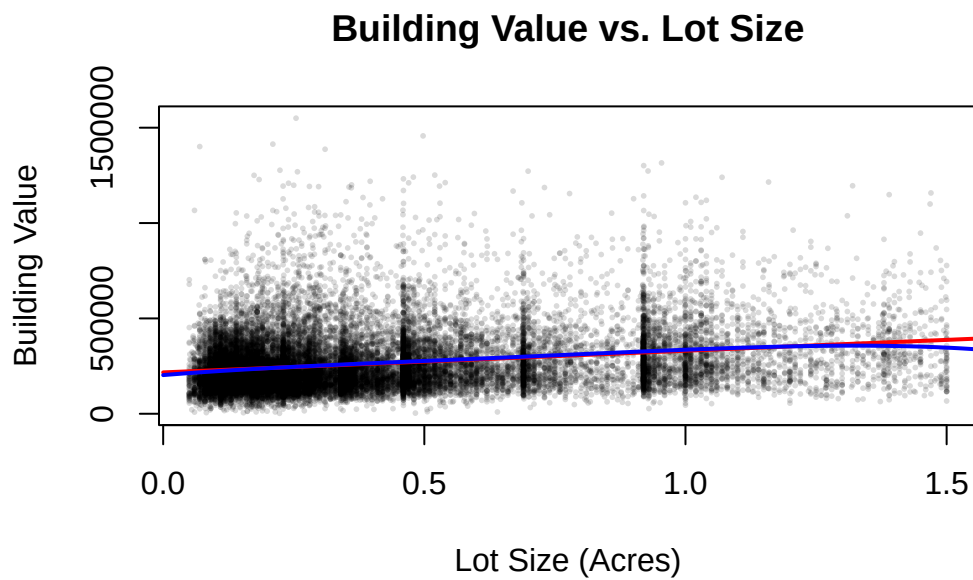
```

    cex = 0.4,
    col = rgb(0, 0, 0, 0.15),
    xlab = "Lot Size (Acres)",
    ylab = "Building Value",
    main = "Building Value vs. Lot Size"
)

lines(
  pred_grid$lot_size_acres,
  pred_grid$pred_building_value,
  col = "red",
  lwd = 2
)

lines(
  pred_grid$lot_size_acres,
  pred_grid$pred_poly,
  col = "blue",
  lwd = 2
)

```



Adding other predictors

Now, let's try to add the number of rooms and the building area to our polynomial model. Let's create indicator variables for each value of `n_rooms` using the `i` function. Control for `building_area` linearly.

```
rooms_poly_mod <- feols(
  building_value ~ poly(lot_size_acres, 4) + i(n_rooms) + building_area,
  data = df,
  vcov = "HC1"
)

summary(rooms_poly_mod)
```

OLS estimation, Dep. Var.: building_value

Observations: 1,655,254

Standard-errors: Heteroskedasticity-robust

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	51595.0843	1.360764e+03	37.91627	< 2.2e-16	***
poly(lot_size_acres, 4)1	6386918.1915	1.328608e+05	48.07226	< 2.2e-16	***
poly(lot_size_acres, 4)2	740874.9433	1.206194e+05	6.14225	8.1378e-10	***
poly(lot_size_acres, 4)3	-1787343.1260	1.224276e+05	-14.59919	< 2.2e-16	***
poly(lot_size_acres, 4)4	-1497824.7462	1.207993e+05	-12.39928	< 2.2e-16	***
n_rooms::4	9622.6650	1.392121e+03	6.91223	4.7726e-12	***
n_rooms::5	21354.1435	1.361601e+03	15.68312	< 2.2e-16	***
n_rooms::6	41048.3157	1.359769e+03	30.18771	< 2.2e-16	***
n_rooms::7	57785.5662	1.366907e+03	42.27470	< 2.2e-16	***
n_rooms::8	93038.2583	1.379095e+03	67.46325	< 2.2e-16	***
n_rooms::9	134106.8826	1.423430e+03	94.21389	< 2.2e-16	***
n_rooms::10	148723.7840	1.463698e+03	101.60824	< 2.2e-16	***
building_area	57.1099	1.195430e-01	477.73464	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 110,862.8 Adj. R2: 0.456179

Answer the following questions:

1. What is the omitted category for `n_rooms`?

`n_rooms::3` is the omitted category, meaning that the coefficients on the other `n_rooms` indicators are interpreted relative to homes with 3 rooms.

2. Interpret in words the coefficient on `n_rooms::7` and comment on its statistical significance.

The coefficient on `n_rooms::7` is \$57,785.60, meaning that homes with 7 rooms are estimated to be worth \$57,785.60 more than homes with 3 rooms, holding lot size and building area constant (*ceteris paribus*). This result is statistically significant at the 1% level.

3. We could have run a regression controlling linearly for the number of rooms (treating it as a continuous measure). Do your estimates for `i(n_rooms)` make you comfortable with making that simplifying assumption?

No, the relationship between the number of rooms and building value does not appear to be linear. The coefficients on `i(n_rooms)` do not increase by a constant amount as the number of rooms increases, which suggests that treating `n_rooms` as a continuous variable would not capture the true relationship well.

4. What is the estimated return on square footage for homes in Massachusetts? To help put this result in context, multiply this coefficient by the standard deviation of `building_area`. Your interpretation will be a for “1 standard deviation increase in `building_area`”

```
sd(df$building_area, na.rm = TRUE)
```

```
[1] 1286.486
```

```
coef(rooms_poly_mod)["building_area"]
```

```
building_area  
57.10987
```

```
coef(rooms_poly_mod)["building_area"] * sd(df$building_area, na.rm = TRUE)
```

```
building_area  
73471.02
```

Answer:

A one standard deviation increase in building area is associated with about a \$73,471 increase in building value, holding lot size and room indicators fixed.

Fixed Effects

One concern we might have with interpreting our regression results is that the number of rooms a house has is likely related to what town the home is in. Let's add a set of town indicators to limit our regression to compare homes "in the same town", so to speak. It turns out there are 98 different towns in our sample, so this will end up slowing down our regression quite a bit. Instead, we will specify the `town` variable after a vertical bar, `|`, in our formula. It will look like `y ~ x | town`. `feols` will estimate these effects very quickly, but note their coefficients are not displayed in the summary.

Modify the previous regression model to add town "fixed effects" (i.e. a set of indicators for each town).

```
rooms_poly_fe_mod <- feols(
  building_value ~ poly(lot_size_acres, 4) + i(n_rooms) + building_area | town,
  data = df,
  vcov = "HC1"
)
```

NOTE: 1 fixed-effect singleton was removed (1 observation).

```
summary(rooms_poly_fe_mod)
```

OLS estimation, Dep. Var.: building_value

Observations: 1,655,253

Fixed-effects: town: 97

Standard-errors: Heteroskedasticity-robust

	Estimate	Std. Error	t value	Pr(> t)	
poly(lot_size_acres, 4)1	16746161.5828	1.570483e+05	106.63061	< 2.2e-16	***
poly(lot_size_acres, 4)2	-7020777.1943	1.217867e+05	-57.64813	< 2.2e-16	***
poly(lot_size_acres, 4)3	1819216.5472	1.141285e+05	15.94007	< 2.2e-16	***
poly(lot_size_acres, 4)4	-1874510.8306	1.096651e+05	-17.09305	< 2.2e-16	***
n_rooms::4	-3786.2697	1.356343e+03	-2.79153	5.2460e-03	**
n_rooms::5	-3360.0376	1.330890e+03	-2.52465	1.1581e-02	*
n_rooms::6	1874.4699	1.331794e+03	1.40748	1.5929e-01	
n_rooms::7	9760.1526	1.339431e+03	7.28679	3.1756e-13	***
n_rooms::8	34766.0283	1.351474e+03	25.72453	< 2.2e-16	***
n_rooms::9	60125.7608	1.388909e+03	43.28992	< 2.2e-16	***
n_rooms::10	64188.9635	1.425896e+03	45.01658	< 2.2e-16	***
building_area	83.6912	1.554260e-01	538.46278	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 95,521.2 Adj. R2: 0.596248
Within R2: 0.524024

How does this result change our estimate on `building_area`?

```
coef(rooms_poly_mod)["building_area"]
```

```
building_area  
57.10987
```

```
coef(rooms_poly_fe_mod)["building_area"]
```

```
building_area  
83.69125
```

Answer:

The estimate for `building_area` is slightly different when we include town fixed effects. The coefficient is now \$83.69, which is higher than the previous estimate of \$57.11.

Zoning Regulation

Zoning regulations limit the kind of development that can occur on a parcel. We have a bunch of zoning measures including: - `z_mls_acres`, the minimum lot size (the required amount of land needed to build a house). - `z_permit_2fam`, whether 2 family homes are allowed to be built “By Right”, “By Permit”, or “Not Allowed”. - `z_residential/z_nonresidential/z_mixed` that says what kind of development can be built on a parcel: residential, commercial, or both.

Using regression methods, try to present a case that more stringent regulations (higher minimum lot size) is associated with larger observed lot sizes. You should write up your results in a paragraph and include a figure. Make sure to interpret the statistical significance of your results.

Answer:

```
zoning_mod <- feols(  
  lot_size_acres ~ z_mls_acres + z_mls_acres^2 + z_mls_acres^3,  
  data = df,  
  vcov = "HC1"  
)  
  
summary(zoning_mod)
```

```

OLS estimation, Dep. Var.: lot_size_acres
Observations: 1,655,254
Standard-errors: Heteroskedasticity-robust

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.004542	0.000749	-6.06485	1.3211e-09 ***
z_mls_acres	0.999723	0.004652	214.88987	< 2.2e-16 ***
I(z_mls_acres^2)	-0.128433	0.006733	-19.07639	< 2.2e-16 ***
I(z_mls_acres^3)	-0.099777	0.002494	-39.99986	< 2.2e-16 ***

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.230352   Adj. R2: 0.472316

```

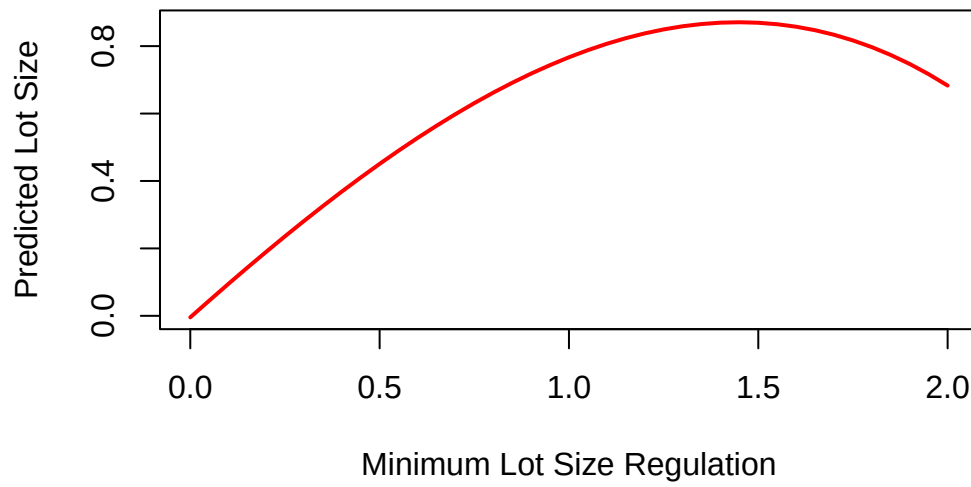
```

zoning_grid <- data.frame(
  z_mls_acres = seq(0, 2, 0.05)
)

zoning_grid$pred_lot_size <- predict(zoning_mod, newdata = zoning_grid)

plot(
  pred_lot_size ~ z_mls_acres,
  data = zoning_grid,
  type = "l",
  col = "red",
  lwd = 2,
  xlab = "Minimum Lot Size Regulation",
  ylab = "Predicted Lot Size"
)

```



The regression shows a positive and statistically significant relationship between stricter minimum lot size regulation and larger observed lot sizes. However, the predicted lot size is always below the minimum lot size regulation.